

VITAMINS



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OBJECTIVES

- At the end of the lecture student should be able to:
- Define vitamins.
- Describe the classification of vitamins.
- Explain the different essential functions performed by vitamins in human body.



VITAMINS

- “abbreviated from vital amines”
- “A vitamin is defined as a naturally occurring essential organic constituent of the diet which in minute amounts aids in maintaining the normal metabolic activities of the tissues”.



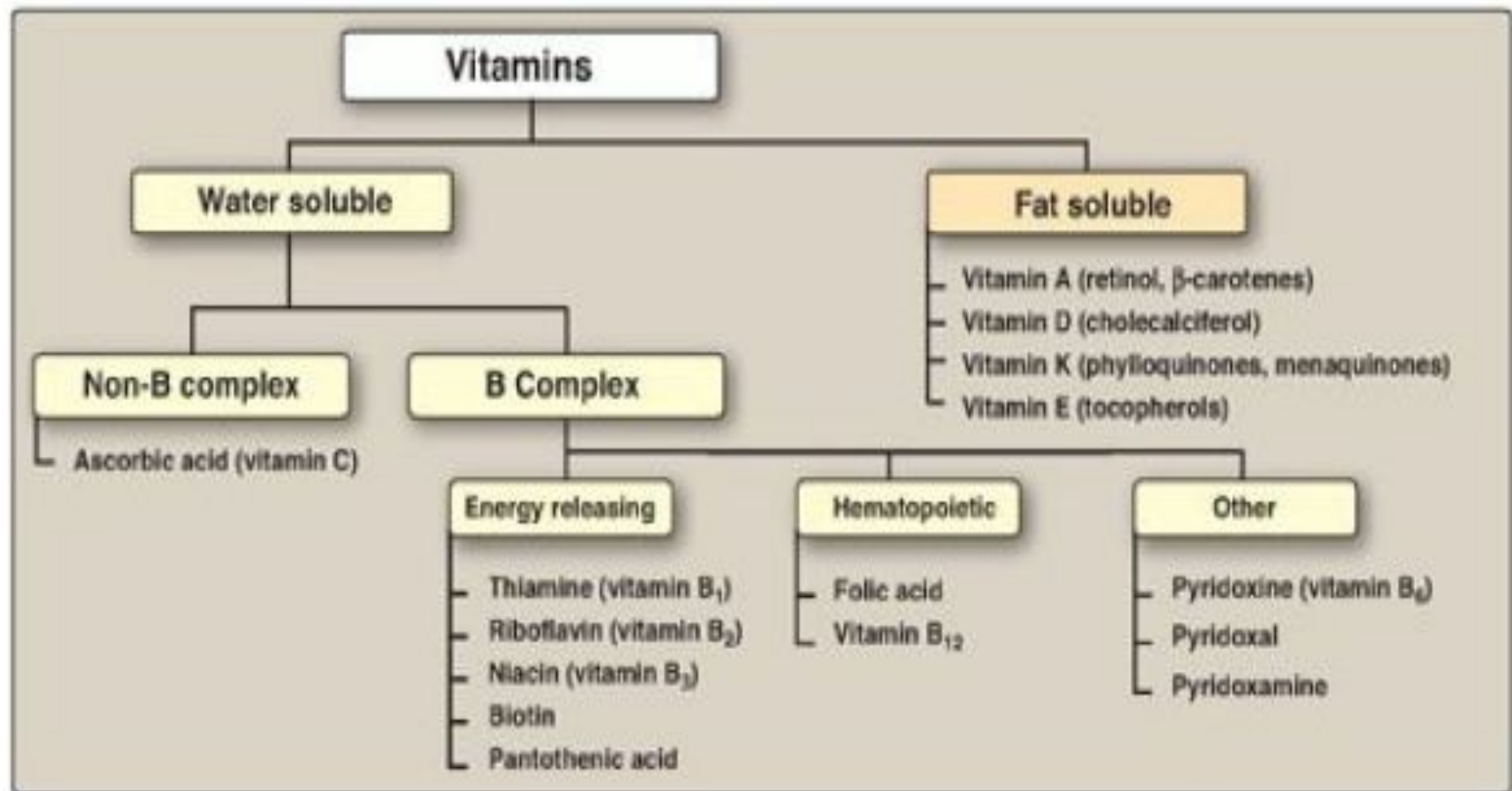
I. OVERVIEW

Vitamins are chemically unrelated organic compounds that cannot be synthesized in adequate quantities by humans and, therefore, must be supplied by the diet. Nine vitamins (folic acid, cobalamin, ascorbic acid, pyridoxine, thiamine, niacin, riboflavin, biotin, and pantothenic acid) are classified as water soluble. Because they are readily excreted in the urine, toxicity is rare. However, deficiencies can occur quickly. Four vitamins (A, D, K, and E) are termed fat soluble . They are released, absorbed, and transported (in chylomicrons) with dietary fat. They are not readily excreted, and significant quantities are stored in the liver and adipose tissue. In fact, consumption of vitamins A and D in excess of the Dietary Reference Intakes can lead to accumulation of toxic quantities of these compounds. Vitamins are required to perform specific cellular functions. For example, many of the water-soluble vitamins are precursors of coenzymes for the enzymes of intermediary metabolism. In contrast to the water-soluble vitamins, only one fat-soluble vitamin (vitamin K) has a coenzyme function.

CLASSIFICATION OF VITAMINS

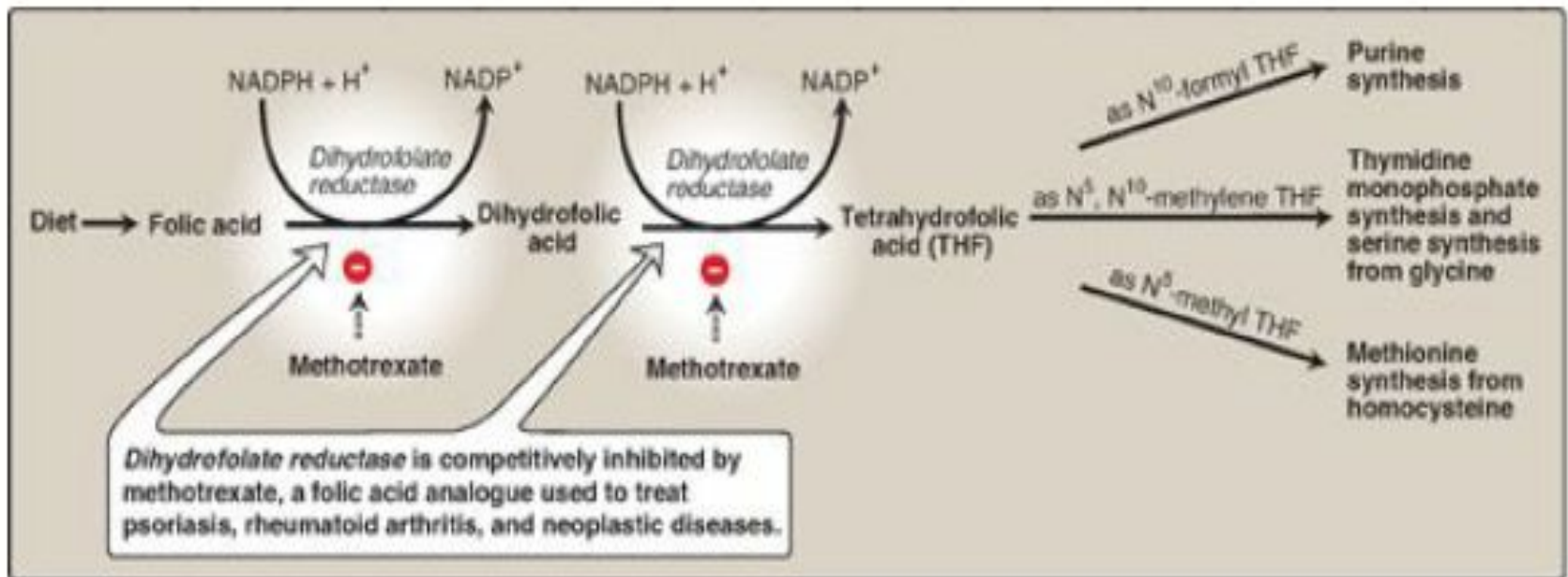
- They can be classified according to their solubility into two groups.
- A: FAT SOLUBLE
- B: WATER SOLUBLE





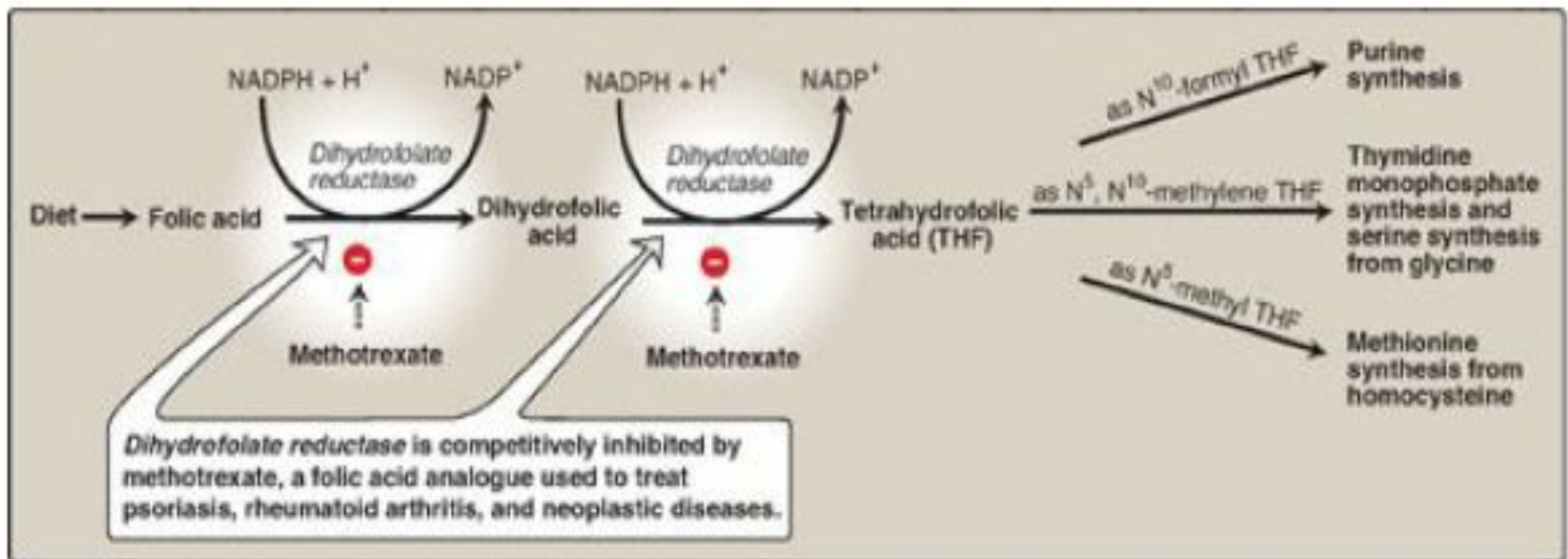
FOLIC ACID

Folic acid (or folate), which plays a key role in one-carbon metabolism, is essential for the biosynthesis of several compounds. Folic acid deficiency is probably the most common vitamin deficiency in the United States, particularly among pregnant women and alcoholics. [Note: Leafy, dark green vegetables are a good source of folic acid.]



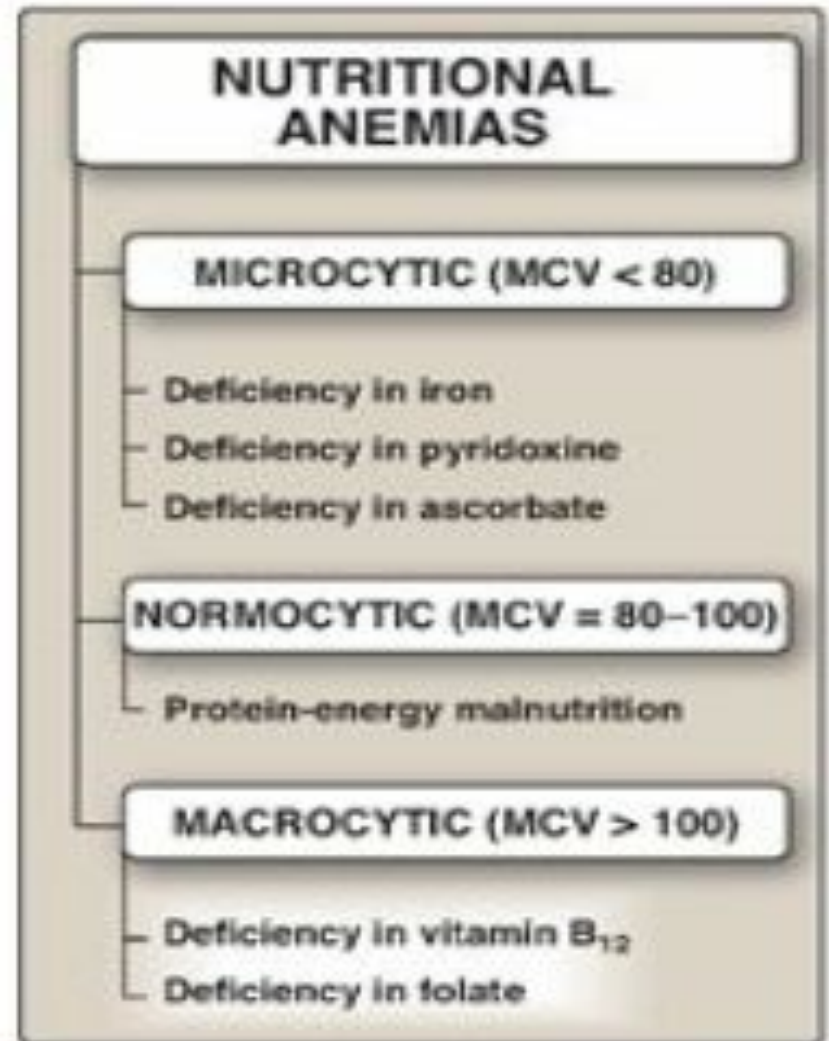
FUNCTION OF FOLIC ACID

- Tetrahydrofolate (THF), the reduced, coenzyme form of folate, receives one-carbon fragments from donors such as serine, glycine, and histidine and transfers them to intermediates in the synthesis of amino acids, purines, and thymidine monophosphate (TMP), a pyrimidine nucleotide found in DNA

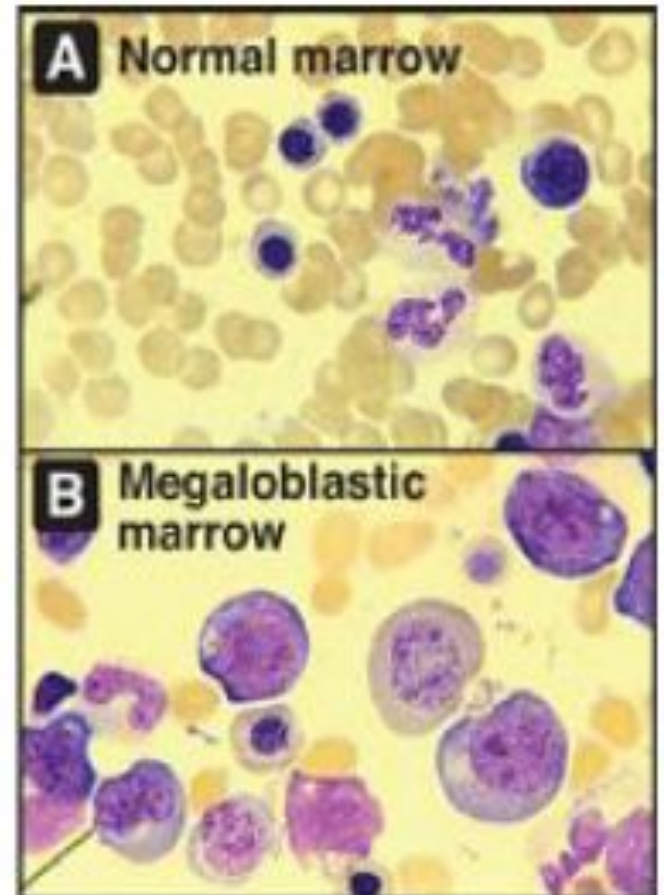
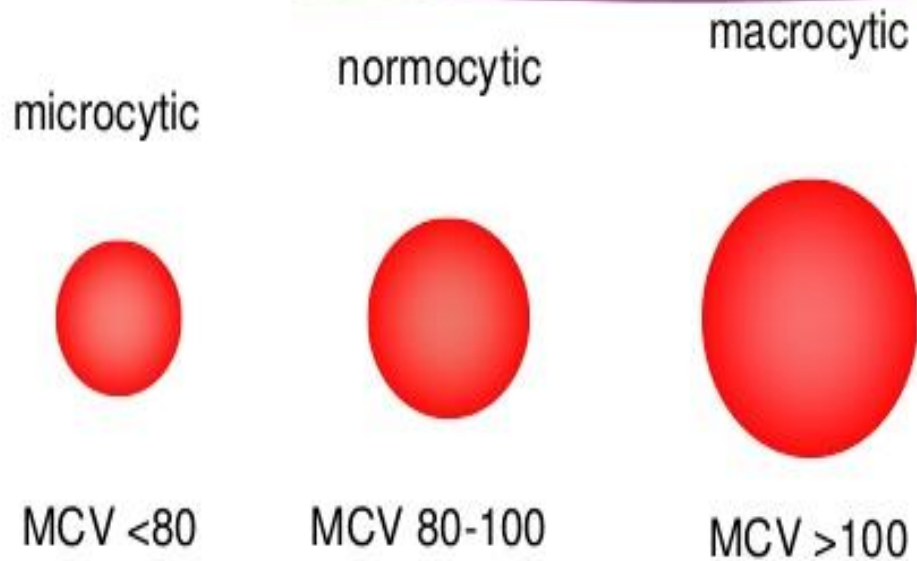


NUTRITIONAL ANEMIAS

- These macrocytic anemias are commonly called megaloblastic because a deficiency of either vitamin (or both) causes accumulation of large, immature RBC precursors, known as megaloblasts, in the bone marrow and the blood.

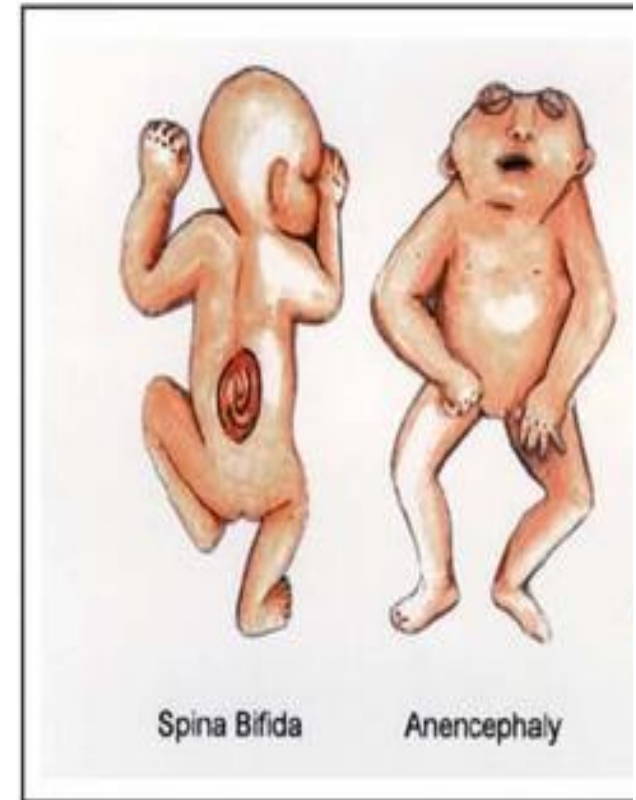


Morphologic classification :



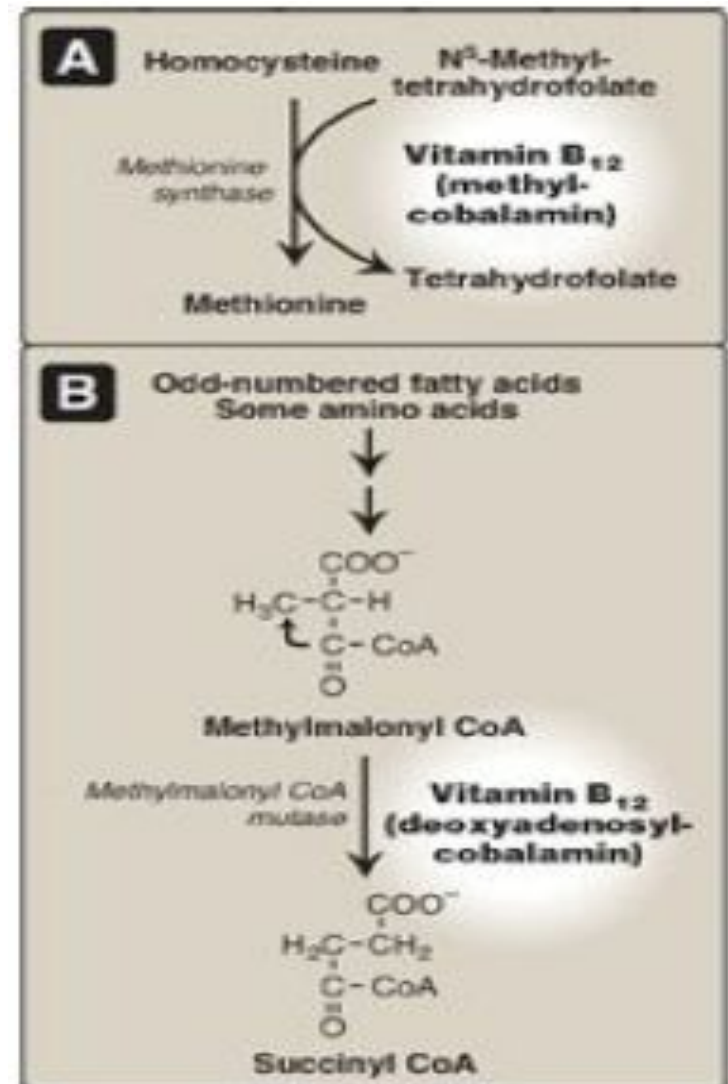
FOLATE AND NEURAL TUBE DEFECTS IN THE FETUS:

- ❑ **Spina bifida and anencephaly**, the most common neural tube defects (NTDs).
- ❑ Folic acid supplementation before conception and during the first trimester has been shown to significantly reduce NTDs.
- ❑ Therefore, all women of childbearing age are advised to consume folic acid to reduce the risk of having a pregnancy affected by NTDs.



COBALAMIN (VITAMIN B12)

- Vitamin B12 is required in humans for two essential enzymatic reactions: the remethylation of homocysteine (Hcy) to methionine and the isomerization of methylmalonyl coenzyme A (CoA), which is produced during the degradation of some amino acids (isoleucine, valine, threonine, and methionine) and fatty acids (FAs) with odd numbers of carbon atoms.



CONT..

- When cobalamin is deficient, unusual (branched) FAs accumulate and become incorporated into cell membranes, including those of the central nervous system (CNS). This may account for some of the neurologic manifestations of vitamin B12 deficiency.
- Folic acid (as N 5-methyl THF) is also required in the remethylation of Hcy.
- Therefore, deficiency of B 12 or folate results in elevated Hcy levels.

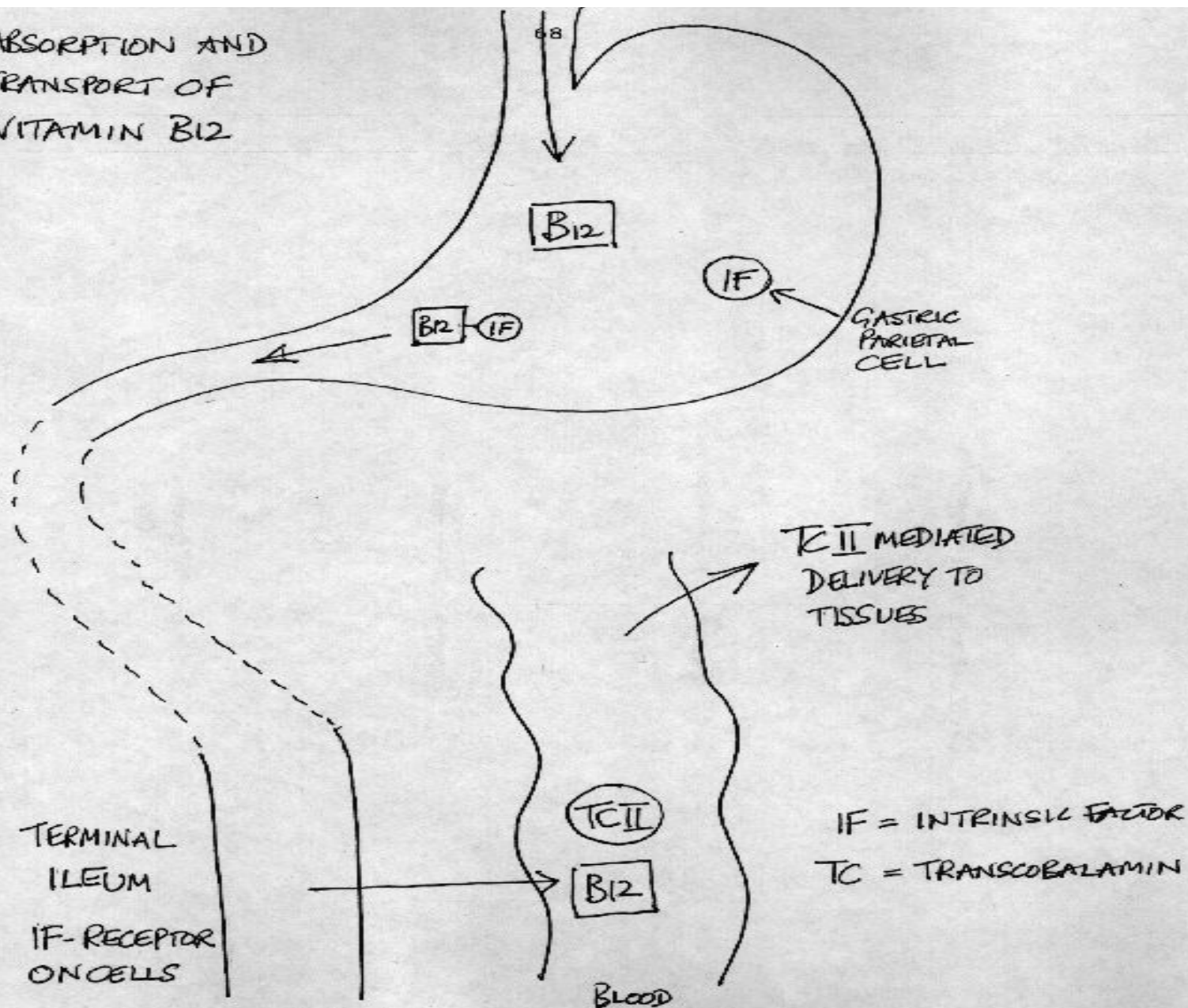


CLINICAL INDICATIONS FOR VITAMIN B12

- ❑ In contrast to other water-soluble vitamins, significant amounts (2–5 mg) of vitamin B12 are stored in the body.
- ❑ As a result, it may take several years for the clinical symptoms of B12 deficiency to develop as a result of decreased intake of the vitamin.
- ❑ Deficiency happens much more quickly if absorption is impaired



ABSORPTION AND TRANSPORT OF VITAMIN B₁₂



VITAMIN B12 DEFICIENCY:

A. Pernicious anemia:

- Most common form.
- It is considered to be an autoimmune disorder.
- Causes: autoimmune gastritis associated with failure of production of intrinsic factor, essential for vitamin B12 absorption. The chronic gastritis is also associated with **Achlorhydria, anti-intrinsic factor and antiparital cell antibodies with increased incidence of gastric carcinoma.**



B. OTHER FORMS OF VITAMIN B12 DEFICIENCY

- 1. Total gastric resection**
- 2. Disorder of the distal ileum**
- 3. Strict vegetarian diet**
- 4. Intestinal malabsorption syndrome**
- 5. Diphylobothrium latum infestation; the fish tapeworm, acquired by ingestion of freshwater fish, inhabits the intestine and causes vitamin B12 depletion.**



ASCORBIC ACID (VITAMIN C)

- The active form of vitamin C is **ascorbic acid**.
- The main function of ascorbate is as a reducing agent in several different reactions.
- Vitamin C has a well documented role as a coenzyme in hydroxylation reactions (for example, hydroxylation of collagen).
- Vitamin C is, therefore, required for the maintenance of normal connective tissue as well as for wound healing.
- Vitamin C also reduces ferric iron to the ferrous form, thereby facilitating the absorption of dietary iron from the intestine.



A. DEFICIENCY OF ASCORBIC ACID

- A deficiency of ascorbic acid results in scurvy, a disease characterized by sore and spongy gums, loose teeth, fragile blood vessels, swollen joints, fatigue, and a microcytic anemia caused by decreased absorption of iron.
- Many of the deficiency symptoms can be explained by a deficiency in the hydroxylation of collagen, resulting in defective connective tissue.





B. PREVENTION OF CHRONIC DISEASE

- Vitamin C is one of a group of nutrients that includes vitamin E and β -carotene, which are known as antioxidants.
- Consumption of diets rich in these compounds is associated with a decreased incidence of some chronic diseases, such as coronary heart disease and certain cancers.
- However, clinical trials involving supplementation with the isolated antioxidants have failed to demonstrate any convincing beneficial effects.



B COMPLEX

VITAMIN	OTHER NAMES	ACTIVE FORM	FUNCTION
Vitamin B ₆	Pyridoxine Pyridoxamine Pyridoxal	Pyridoxal phosphate	Coenzyme for enzymes, particularly in amino acid metabolism
Vitamin B ₁	Thiamine	Thiamine pyrophosphate	Coenzyme of enzymes catalyzing: Pyruvate → acetyl CoA α-Ketoglutarate → Succinyl CoA Ribose 5-P + xylulose 5-P → Sedoheptulose 7-P + Glyceraldehyde 3-P Branched-chain α-keto acid oxidation
Niacin	Nicotinic acid Nicotinamide	NAD ⁺ , NADP ⁺	Electron transfer
Vitamin B ₂	Riboflavin	FMN, FAD	Electron transfer
Biotin	—	Enzyme-bound biotin	Carboxylation reactions
Pantothenic acid	—	Coenzyme A	Acyl carrier

VITAMIN	DEFICIENCY	SIGNS AND SYMPTOMS	TOXICITY	NOTES
Vitamin B ₆	Rare	Glossitis Neuropathy	Yes	Deficiency can be induced by isoniazid Sensory neuropathy occurs at high doses
Vitamin B ₁	Beriberi Wernicke-Korsakoff syndrome (most common in alcoholics)	Tachycardia, vomiting, convulsions Apathy, loss of memory, dysregulated eye movements	None	—
Niacin	Pellagra	Dermatitis Diarrhea Dementia	None	High doses of niacin used to treat hyperlipidemia
Vitamin B ₂	Rare	Dermatitis Angular stomatitis	None	—
Biotin	Rare	—	None	Consumption of large amounts of raw egg whites (which contains a protein, avidin, that binds biotin) can induce a biotin deficiency
Pantothenic acid	Rare	—	None	—



VITAMIN A

- The retinol (vitamin A) is essential for:
 - Vision,
 - Reproduction,
 - Growth, and
 - Maintenance of epithelial tissues
- They also play a role in immune function.

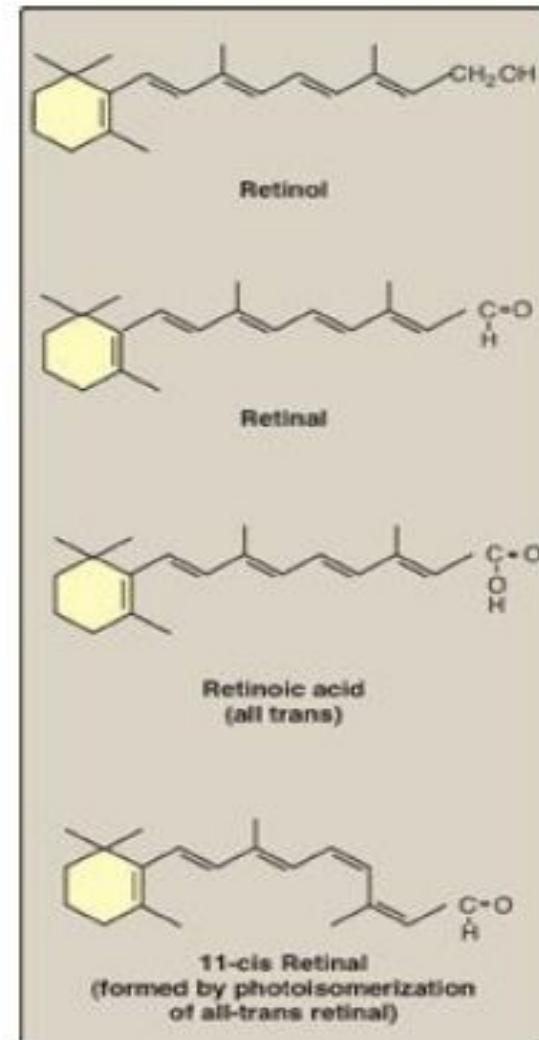


STRUCTURE OF VITAMIN A

- **Vitamin A** is often used as a collective term for several related biologically active molecules.
- The term retinoids includes both natural and synthetic forms of vitamin A that may or may not show vitamin A activity.
- **1. Retinol:** A primary **alcohol**
- **2. Retinal:** This is the **aldehyde** derived from the oxidation of retinol. Retinal and retinol can readily be interconverted.
- **3. Retinoic acid:** This is the **acid** derived from the oxidation of retinal.
- **4. β -Carotene:** Plant foods contain **β -carotene**, which can be oxidatively cleaved in the intestine to yield two molecules of retinal. In humans, the conversion is inefficient, and the vitamin A activity of β -carotene is only about 1/12 that of retinol.

STRUCTURE OF THE RETINOIDS

- Retinoic acid, derived from oxidation of retinol, mediates most of the actions of the retinoids, except for vision, which depends on retinal, the aldehyde derivative of retinol.



ABSORPTION AND TRANSPORT OF VITAMIN A

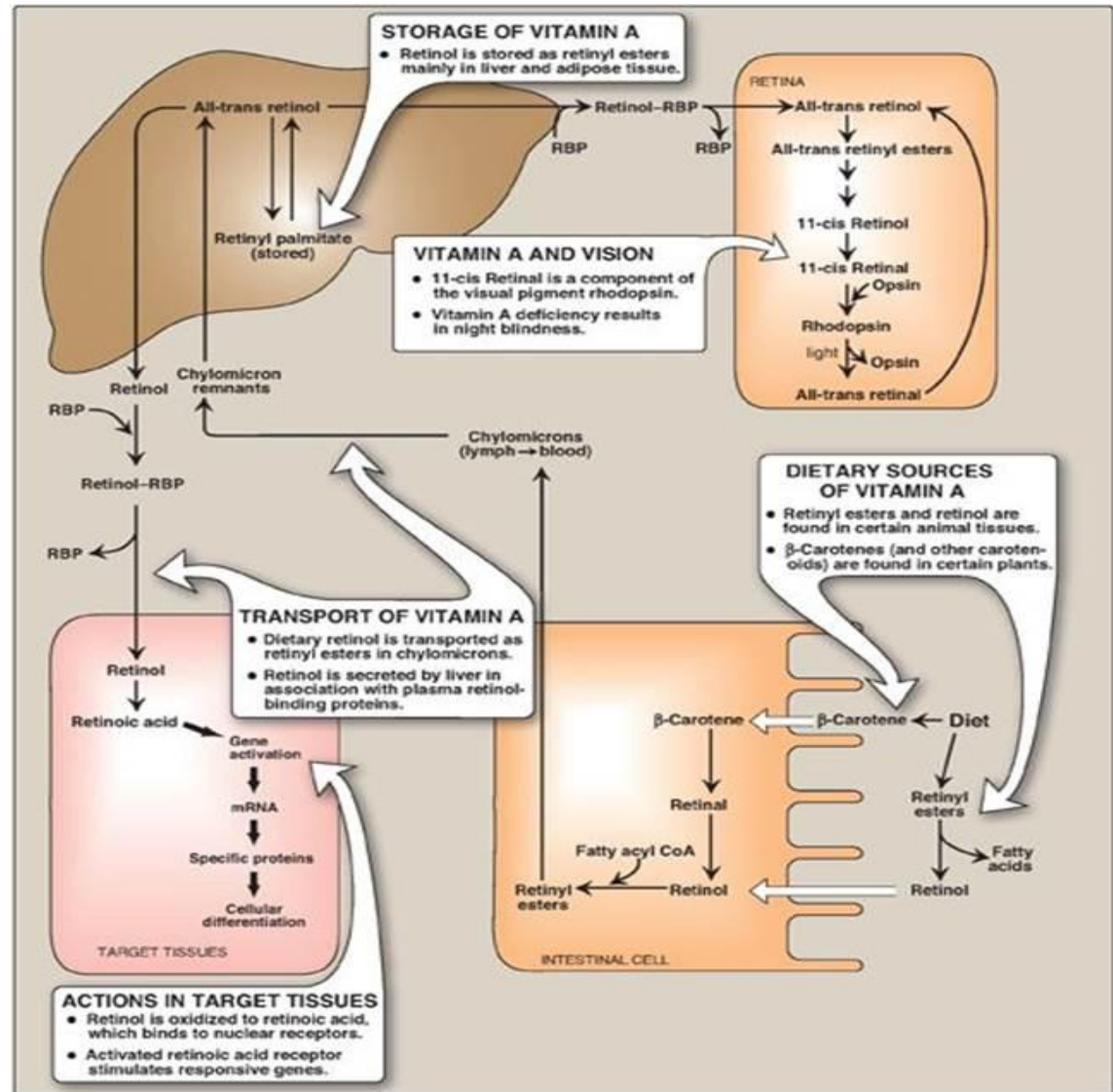
1. Transport to the liver:

- Retinyl esters present in the diet are hydrolyzed in the intestinal mucosa, releasing retinol.
- Retinol is re-esterified to long chain FAs in the intestinal mucosa and secreted as a component of chylomicrons into the lymphatic system.
- Chylomicron remnants are taken up by, and stored in, the liver.



Figure:
 • Absorption, transport, and storage of vitamin A and its derivatives.

- RBP = retinol-binding protein;
- CoA = coenzyme A;
mRNA = messenger RNA.



2. Release from the liver:

- When needed, retinol is released from the liver and transported to extrahepatic tissues by the plasma retinol-binding protein (RBP).
- The retinol–RBP complex binds to a transport protein on the surface of the cells of peripheral tissues, permitting retinol to enter.
- Many tissues contain a cellular retinol binding protein that carries retinol to sites in the nucleus where the vitamin acts in a manner analogous to that of steroid hormones.



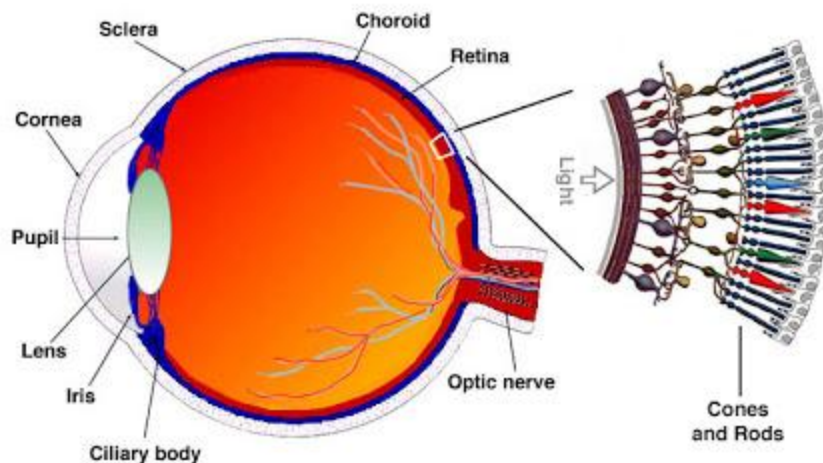
D. FUNCTIONS OF VITAMIN A

1. Visual cycle:

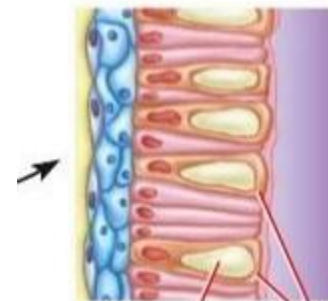
- Vitamin A is a component of the visual pigments of rod and cone cells. Rhodopsin, the visual pigment of the rod cells in the retina.

2. Maintenance of epithelial cells:

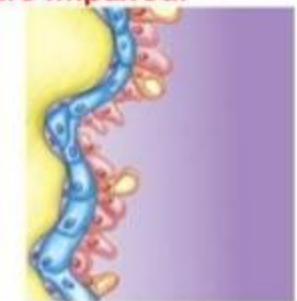
- Vitamin A is essential for normal differentiation of epithelial tissues and mucus secretion, and thus, supports the body's barrier based defense against pathogens.



Vitamin A maintains healthy cells in the mucous membranes.



Without vitamin A, the normal structure and function of the cells in the mucous membranes are impaired.



Mucus Goblet cells

3. Reproduction:

- Retinol and retinal are essential for normal reproduction, supporting spermatogenesis in the male and preventing fetal resorption in the female.
- Retinoic acid is inactive in maintaining reproduction and in the visual cycle but promotes growth and differentiation of epithelial cells.
- Therefore, animals given vitamin A only as retinoic acid from birth are blind and sterile.



DISTRIBUTION OF VITAMIN A

- ❑ Liver, kidney, cream, butter, and egg yolk are good sources of preformed vitamin A.
- ❑ Yellow, orange, and dark green vegetables and fruits are good dietary sources of the carotenes, which serve as precursors of vitamin A.



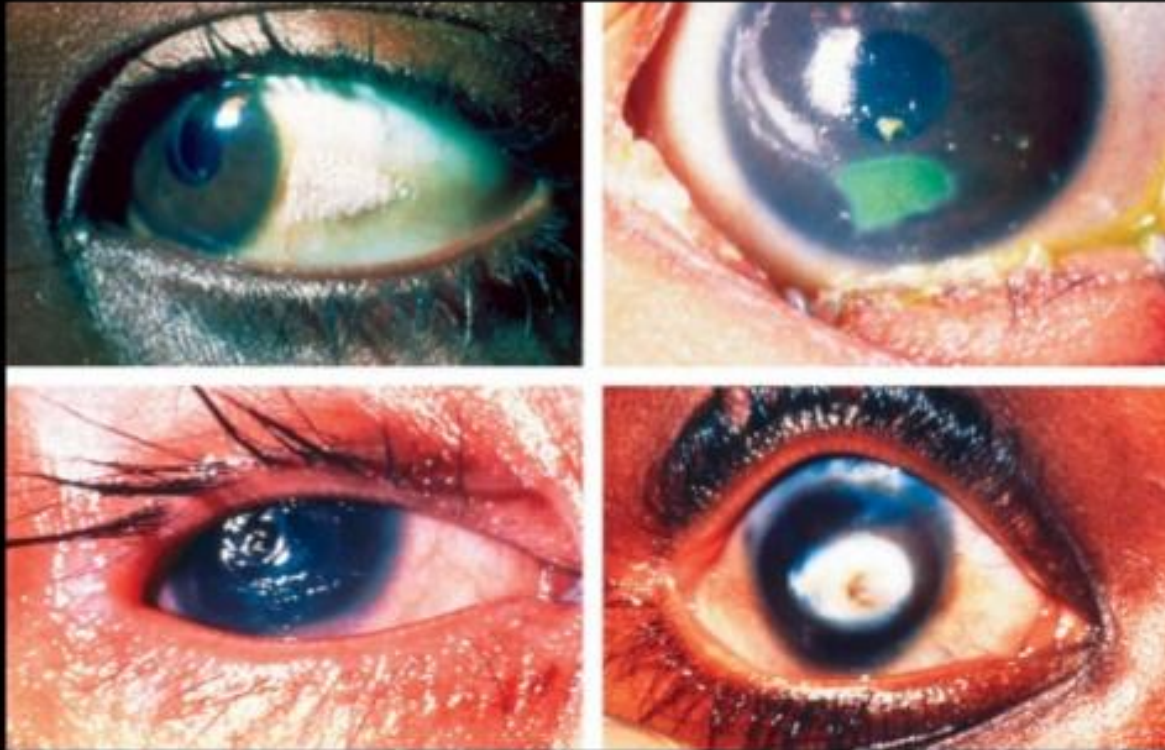
CLINICAL INDICATIONS

- Retinol and its carotenoid precursor are used as dietary supplements, whereas various forms of retinoic acid are useful in dermatology.

1. Dietary deficiency:

- Vitamin A, administered as retinol or retinyl esters, is used to treat patients who are deficient in the vitamin .
 - Night blindness is one of the earliest signs of vitamin A deficiency.
 - Severe vitamin A deficiency leads to xerophthalmia, a pathologic dryness of the conjunctiva and cornea, caused, in part, by increased keratin synthesis. If untreated, xerophthalmia results in corneal ulceration and, ultimately, in blindness because of the formation of opaque scar tissue.
 - The condition is most commonly seen in children in developing countries.
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Signs of Xerophthalmia



Top left: Bitot's spot

Top right: Cornea with typical dry appearance and ulcer stained green with fluorescein dye;

Bottom right: Corneal scarring;

Bottom left: Deep corneal ulceration progressing towards keratomalacia.

2. Acne and psoriasis:

- Dermatologic problems such as acne and psoriasis are effectively treated with retinoic acid or its derivatives.
- Tretinoin is too toxic for systemic administration and is confined to topical application.



TOXICITY OF RETINOIDS

Vitamin A:

- ❑ Excessive intake of vitamin A produces a toxic syndrome called hypervitaminosis A.
- ❑ Early signs of chronic hypervitaminosis A are reflected in the skin, which becomes dry and pruritic (due to decreased keratin synthesis); the liver, which becomes enlarged and can become cirrhotic; and in the CNS, where a rise in intracranial pressure may mimic the symptoms of a brain tumor.
- ❑ Pregnant women particularly should not ingest excessive quantities of vitamin A because of its potential for teratogenesis.



VITAMIN D

- The D vitamins are a group of sterols that have a hormone-like function.
- The active molecule, 1,25-dihydroxycholecalciferol ([1,25-diOH-D3] calcitriol), binds to intracellular receptor proteins.
- The 1,25-diOH-D3–receptor complex interacts with DNA in the nucleus of target cells in a manner similar to that of vitamin A.
- The most prominent actions of 1,25-diOH-D3 are to regulate the plasma levels of calcium and phosphorus.



DISTRIBUTION OF VITAMIN D

1. Endogenous vitamin precursor:

- 7-Dehydrocholesterol, an intermediate in cholesterol synthesis, is converted to cholecalciferol in the dermis and epidermis of humans exposed to sunlight and transported to liver bound to vitamin D-binding protein.

2. Diet:

- Ergocalciferol (vitamin D₂), found in plants, and cholecalciferol (vitamin D₃), found in animal tissues, are sources of preformed vitamin D activity.
- Dietary vitamin D is packaged into chylomicrons.
- [Note: Preformed vitamin D is a dietary requirement only in individuals with limited exposure to sunlight.]



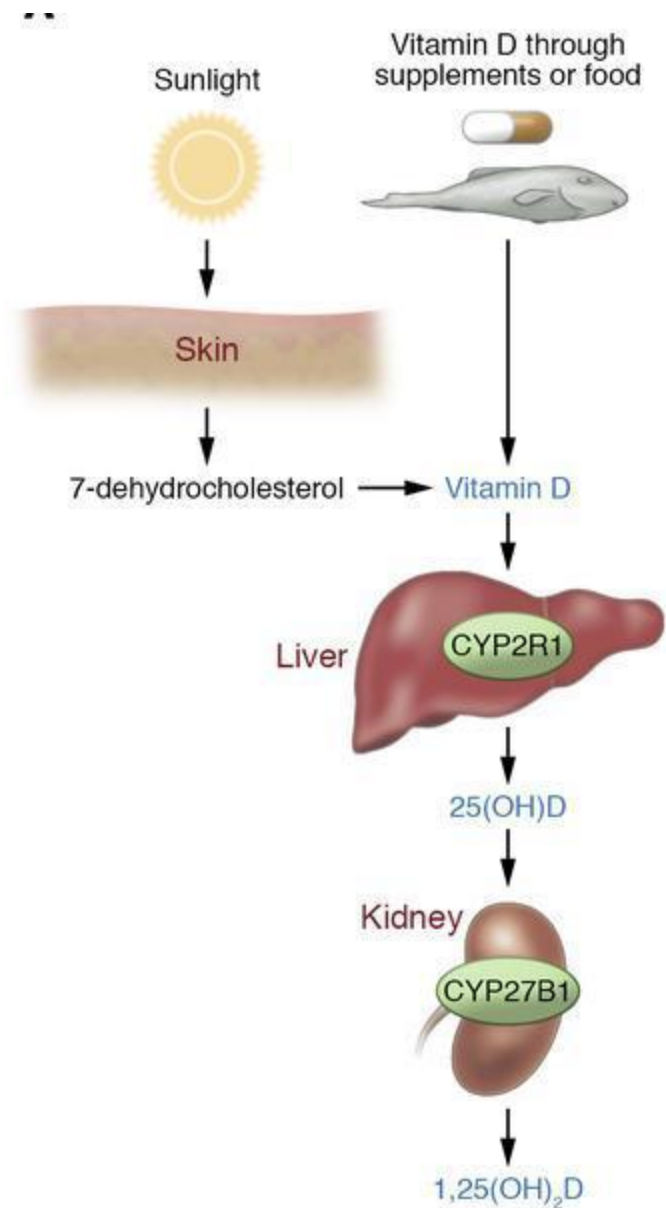
METABOLISM OF VITAMIN D

1. Formation of 1,25-dihydroxycholecalciferol:

- Vitamins D2 and D3 are not biologically active but are converted in vivo to the active form of the D vitamin by two sequential hydroxylation reactions.
- The first hydroxylation occurs at the 25 position and is catalyzed by a specific 25-hydroxylase in the liver. The product of the reaction, 25-hydroxycholecalciferol ([25-OH-D3], calcidiol), is the predominant form of vitamin D in the plasma and the major storage form of the vitamin.
- 25-OH-D3 is further hydroxylated at the 1 position by 25-hydroxycholecalciferol 1-hydroxylase found primarily in the kidney, resulting in the formation of 1,25-diOH-D3 (calcitriol).



METABOLISM OF VITAMIN D



FUNCTION OF VITAMIN D

- The overall function of 1,25-diOH-D3 is to maintain adequate plasma levels of calcium.

- It performs this function by:
 - 1) increasing uptake of calcium by the intestine,
 - 2) minimizing loss of calcium by the kidney by increasing reabsorption, and
 - 3) stimulating resorption (demineralization) of bone when blood calcium is low .



EFFECT OF VITAMIN D ON BONE:

- 1,25-diOH-D3 stimulates the mobilization of calcium from bone by a process that requires presence of PTH.
- The result is an increase in plasma calcium and phosphate.
- Therefore, bone is an important reservoir of calcium that can be mobilized to maintain plasma levels.
- PTH and calcitriol also work together to prevent renal loss of calcium.



DISTRIBUTION OF VITAMIN D

- Vitamin D occurs naturally in fatty fish, liver, and egg yolk. Milk, unless it is artificially fortified, is not a good source of the vitamin.
- Because breast milk is a poor source of vitamin D, supplementation is recommended for breastfed babies.
- Note: 1 mg = 40 international units (IUs).

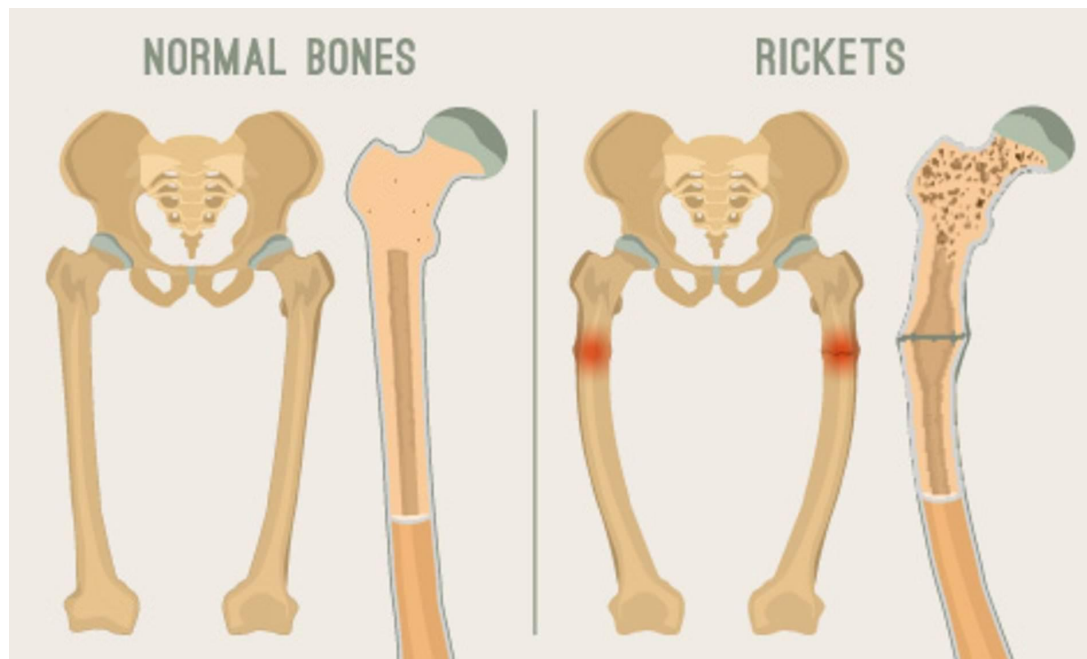
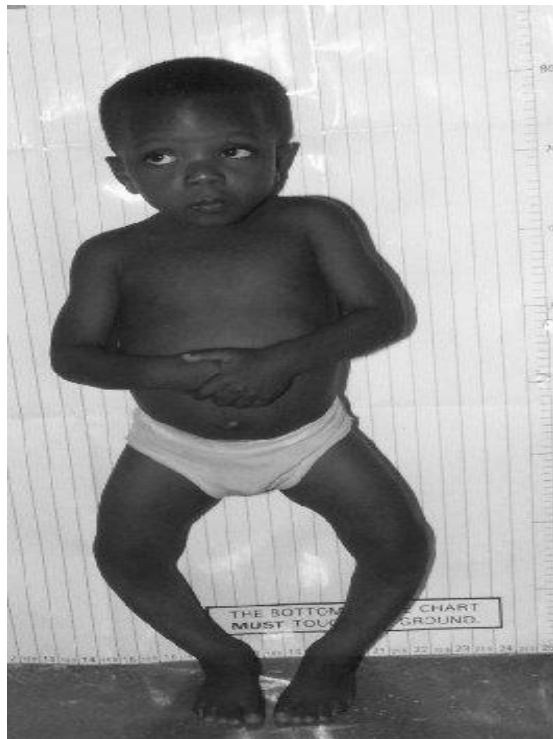


CLINICAL INDICATIONS

❑ 1. Nutritional rickets:

- ❑ Vitamin D deficiency causes a net demineralization of bone, resulting in rickets in children and osteomalacia in adults.
- ❑ Rickets is characterized by the continued formation of the collagen matrix of bone, but incomplete mineralization results in soft, pliable bones.
- ❑ In osteomalacia, demineralization of pre-existing bones increases their susceptibility to fracture.
- ❑ Insufficient exposure to daylight and/or deficiencies in vitamin D consumption occur predominantly in infants and the elderly.
- ❑ Vitamin D deficiency is more common in the northern latitudes, because less vitamin D synthesis occurs in the skin as a result of reduced exposure to ultraviolet light.







RENAL OSTEODYSTROPHY:

- ❑ Chronic kidney disease causes decreased ability to form active vitamin D as well as increased retention of phosphate, resulting in hyperphosphatemia and hypocalcemia.
- ❑ The low blood calcium causes a rise in PTH and associated bone demineralization with release of calcium and phosphate.
- ❑ Supplementation with calcitriol is an effective therapy..



TOXICITY OF VITAMIN D

- ❑ Like all fat-soluble vitamins, vitamin D can be stored in the body and is only slowly metabolized.
- ❑ High doses (100,000 IUs for weeks or months) can cause loss of appetite, nausea, thirst, and stupor.
- ❑ Enhanced calcium absorption and bone resorption results in hypercalcemia, which can lead to deposition of calcium in many organs, particularly the arteries and kidneys.
- ❑ Toxicity is only seen with use of supplements.
- ❑ Excess vitamin D produced in the skin is converted to inactive forms.



